

Table 3 – PHA Study Nodes/Topics Modified

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
1. Node 1: Condensate Feed from Condensate Feed Pumps to Condensate Feed Filter/Coalescer through Preheaters to Naphtha Stabilizer Column (Pumps PM-111A/B, Filter FL-101A/B, Heat Exchanger EX-101/102/103/104/105)	Feed from 200-1/2/3 at ambient temperature; Desalter runs ~175-180 degF, Desalter pressure ~140 psig, 1600 GPM maximum rate. Tie-in to Desalter between downstream of Heat Exchanger EX-104 and upstream of Heat Exchanger EX-105. Coalescers FL-101A/B are currently bypassed. PM-111A/B/C: - 1472 GPM @ 776' TDH FL-101A/B (permanently bypassed): - MAWP = 325 psig @ 120 degF - Capacity = 1460 GPM EX-101: - Shell MAWP = 305 psig @ 450 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 1.34 MMBTU/Hr EX-102: - Shell MAWP = 305 psig @ 450 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 13.6 MMBTU/Hr EX-103: - Shell MAWP = 305 psig @ 460 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 33.8 MMBTU/Hr EX-104: - Shell MAWP = 311 psig @ 650 degF - Tube MAWP = 311 psig @ 650 degF - Duty = 18.9 MMBTU/Hr EX-105: - Shell MAWP = 305 psig @ 650 degF - Tube MAWP = 305 psig @ 650 degF - Duty = 29.7 MMBTU/Hr	40-PI-9004	PM-111A/B/C	Condensate Feed Pumps
			FL-101A/B	Condensate Feed Filter/Coalescer
		40-PI-9004 Redline	PM-111A/B/C	Condensate Feed Pumps
			FL-101A/B	Condensate Feed Filter/Coalescer
		40-PI-9005	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
		40-PI-9005 Redline	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
		40-PI-9006	EX-104	Diesel to Feed Preheater
			EX-105	Bottoms to Feed Preheater
		40-PI-9007	T-101	Naphtha Stabilizer Column
2. Node 2: Naphtha Stabilizer Column, Naphtha Stabilizer Reboiler, Combi Tower Feed Preheater to Combi Tower (T-101, Heat Exchanger EX 106, 107)	Bottoms through reboiler, heated with heat transfer fluid, heat transfer fluid flow controlled by column temperature. Bottoms feed to Combi Tower heated by heat transfer fluid (Heat Exchanger EX-107A/B), heat transfer fluid flow controlled by Combi Tower feed temperature. T-101: - Design = 120 psig MAWP @ 375 degF - MDMT = 10 degF @ 120 psig	40-PI-9007	T-101	Naphtha Stabilizer Column
		40-PI-9010	EX-106	Naphtha Stabilizer Reboiler
			EX-107	COMBI Tower Feed Preheater
		40-PI-9010 Redline	EX-106	Naphtha Stabilizer Reboiler
			EX-107	COMBI Tower Feed Preheater
		40-PI-9014	T-102	COMBI Tower

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
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	EX-106: - Shell MAWP = 287 psig @ 650 degF - Shell MDMT = 287 psig @ 20 degF - Tube MAWP = 276 psig @ 700 degF - Tube MDMT = 276 psig @ 20 degF - Duty = 54.3 MMBTU/Hr			
	EX-107: - Shell MAWP = 227 psig @ 650 degF - Shell MDMT = 227 psig @ 20 degF - Tube MAWP = 218 psig @ 700 degF - Tube MDMT = 218 psig @ 20 degF - Duty = 121.7 MMBTU/Hr			
3. Node 3: Naphtha Stabilizer Column overhead runs at 90-120F, 90 psig, accumulator vents to recycle compressor system system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)	AF-101: - Design = 120 psig MAWP @ 300 degF - Design Duty = 10.7 MMBTU/Hr PV-101: - Design = 120 psig MAWP @ 300 degF - MAWP = 708 psig @ 130 degF - MDMT = 10 degF @ 120 psig PM-101A/B: - GPM = 239, HP = 15, RPM = 3550 - Temp = 120 degF @ deltaP 56 psi	40-PI-9007	T-101	Naphtha Stabilizer Column
		40-PI-9008	AF-101	Naphtha Stabilizer Condenser
		40-PI-9009	PM-101A/B	Naphtha Stabilizer Overhead Pumps
			PV-101	Naphtha Stabilizer Accumulator
		40-PI-9009 Redline	PM-101A/B	Naphtha Stabilizer Overhead Pumps
			PV-101	Naphtha Stabilizer Accumulator
4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)	Combi Tower Bottoms - 525-560F, 9-11 psig, temperature to storage = 140-160F, product flow rate = 250-450 GPM; bottoms product temperature controlled by turning fin fans on/off EX-101: - Shell MAWP = 305 psig @ 450 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 1.34 MMBTU/Hr AF-102: - Design = 50 psig MAWP @ 300 degF - Design Duty = 150.5 MMBTU/Hr PV-102: - Design = 50 psig MAWP @ 300 degF - MDMT = 10 degF @ 50 psig PM-103A/B: - MAWP = 533 psig @ 139 degF - Design = 121 degF @ 119 psi deltaP	40-PI-9006	EX-104	Diesel to Feed Preheater
			EX-105	Bottoms to Feed Preheater
		40-PI-9007	T-101	Naphtha Stabilizer Column
		40-PI-9014	T-102	COMBI Tower
		40-PI-9025	PM-104A/B	COMBI Tower Bottoms Product Pumps
		40-PI-9026	AF-103	Bottoms Product Cooler
		40-PI-9059A	EX-212A/B	Bottoms Product to Desalter Wash Water Heater
			EX-112A/B	Bottoms Product to Desalter Wash Water Heater
		40-PI-9005	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
		40-PI-9005 Redline	EX-101	Light Naphtha to Feed Preheater

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
5. Node 5: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-102, PV-102, PM-103 A/B, Heat Exchanger EX-101) to tanks 24/25	Overhead pressure - 9-11 psig, overhead temperature ~188F, temperature to storage - 67-72F after CH-1/2 chillers. 12 fin fan banks, 24 fans. Typically run all fans, may be able to shut 5 fans in winter. EX-105: - Shell MAWP = 305 psig @ 650 degF - Tube MAWP = 305 psig @ 650 degF - Duty = 29.7 MMBTU/Hr T-102: - Design = 50 psig MAWP @ 575 degF - MDMT = 50 psig @ -20 degF PM-104A/B: - MAWP = 562 psig @ 575 degF @ deltaP 105 psi - Design = 631 GPM, 75 HP, 3550 RPM, 548 degF AF-103: - Design = 250 psig MAWP @ 625 degF - Design Duty = 49.5 MMBTU/Hr	40-PI-9005	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
		40-PI-9005 Redline	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
		40-PI-9014	T-102	COMBI Tower
		40-PI-9015	AF-102	COMBI Tower Condenser
		40-PI-9016	AF-102	COMBI Tower Condenser
		40-PI-9017	AF-102	COMBI Tower Condenser
		40-PI-9018	PM-103A/B	COMBI Tower Overhead Pumps
			PV-102	COMBI Tower Accumulator
		40-PI-9022	N/A	Light Naphtha Product to Storage
		40-PI-9022 Redline	N/A	Light Naphtha Product to Storage
6. Node 6: Heavy Naphtha Side Draw From T-102 to Heavy Naphtha Stripper (T-103), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-102 (T-103, Heat Exchanger EX-108, PM-105A/B, Heat Exchanger EX-102, AF-105)	Normally run with no hot oil on reboiler (Heat Exchanger EX-108), cooled to ~90F in AF-105, stripper pressure approximately same as Combi Tower. EX-102: - Shell MAWP = 305 psig @ 450 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 13.6 MMBTU/Hr EX-108: - Shell MAWP = 83 psig @ 650 degF - Shell MDMT = -20 degF @ 83 psig - Tube MAWP = 156 psig @ 700 degF - Tube MDMT = -20 degF @ 156 psig - Duty = 13.4 MMBTU/Hr T-103: - Design = 50 psig MAWP @ 400 degF - MDMT = 10 degF @ 50 psig PM-105A/B: - MAWP = 642 psig @ 350 degF	40-PI-9005	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
		40-PI-9005 Redline	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
		40-PI-9014	T-102	COMBI Tower
		40-PI-9019	T-103	Heavy Naphtha Stripper
			EX-108	Heavy Naphtha Stripper Reboiler
		40-PI-9021	PM-105A/B	Heavy Naphtha Stripper Product Pumps
			PM-106A/B	Keto Stripper Product Pumps
			PM-107A/B	Diesel Stripper Product Pumps
		40-PI-9022	N/A	Light Naphtha Product to Storage

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
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	- Design = 493 GPM, 60 HP, 3550 RPM, 314 degF AF-105: - Design = 250 psig MAWP @ 375 degF - Design Duty = 15.3 MMBTU/Hr	40-PI-9022 Redline	N/A	Light Naphtha Product to Storage
		40-PI-9023	AF-105	Heavy Naphtha Product Cooler
7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)	Kero stripper runs at ~360-390F, stripper pressure approx. same as Combi Tower. Kero Filters - moisture and particulate removal	40-PI-9005	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
	EX-103: - Shell MAWP = 305 psig @ 450 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 33.8 MMBTU/Hr	40-PI-9005 Redline	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
	EX-109: - Shell MAWP = 258 psig @ 650 degF - Shell MDMT = -20 degF @ 258 psig - Tube MAWP = 298 psig @ 700 degF - Tube MDMT = -20 degF @ 298 psig - Duty = 24.7 MMBTU/Hr	40-PI-9014	T-102	COMBI Tower
		40-PI-9020	T-104	Kero Stripper
			EX-109	Kero Stripper Reboiler
	T-104: - Design = 50 psig MAWP @ 600 degF - MDMT = 10 degF @ 50 psig	40-PI-9021	PM-105A/B	Heavy Naphtha Stripper Product Pumps
			PM-106A/B	Keto Stripper Product Pumps
			PM-107A/B	Diesel Stripper Product Pumps
	PM-106A/B: - MAWP = 535 psig @ 450 degF - Design = 978 GPM, 100 HP, 1775 RPM, 412 degF	40-PI-9022	N/A	Light Naphtha Product to Storage
		40-PI-9022 Redline	N/A	Light Naphtha Product to Storage
		40-PI-9024	AF-104	Kero Product Cooler
	AF-104: - Design = 250 psig MAWP @ 450 degF - Design Duty = 46.4 MMBTU/Hr			
8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)	Diesel stripper runs at ~525-530F, stripper pressure approx. same as Combi Tower	40-PI-9006	EX-104	Diesel to Feed Preheater
			EX-105	Bottoms to Feed Preheater
	EX-104: - Shell MAWP = 311 psig @ 650 degF - Tube MAWP = 311 psig @ 650 degF - Duty = 18.9 MMBTU/Hr	40-PI-9014	T-102	COMBI Tower
		40-PI-9021	PM-105A/B	Heavy Naphtha Stripper Product Pumps
			PM-106A/B	Keto Stripper Product Pumps
	PM-107A/B: - MAWP = 543 psig @ 650 degF - Design = 320 GPM, 50 HP, 3575 RPM, 608 degF	40-PI-9022	PM-107A/B	Diesel Stripper Product Pumps
			N/A	Light Naphtha Product to Storage

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
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	EX-110: - Shell MAWP = 96 psig @ 650 degF - Shell MDMT = -20 degF @ 96 psig - Tube MAWP = 186 psig @ 700 degF - Tube MDMT = -20 degF @ 186 psig - Duty = 7.04 MMBTU/Hr	40-PI-9022 Redline	N/A	Light Naphtha Product to Storage
	AF-106: - MAWP = 250 psig @ 700 degF - Design Duty = 26.14 MMBTU/Hr	40-PI-9025	PM-104A/B	COMBI Tower Bottoms Product Pumps
		40-PI-9026	AF-103	Bottoms Product Cooler
		40-PI-9049	T-105	Diesel Stripper
			EX-110	Diesel Stripper Reboiler
		40-PI-9050	AF-106	Diesel Product Cooler
9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)	Inlet pressure 8-12 psig to ~70 psig through compressor, liquids knocked out and returned to Stabilizer feed CM-101A/B: - Suction P = 16 psia @ 165 degF - Discharge P = 80 psia @ 205 degF	40-PI-9005	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
	PV-108: - MAWP = 150 psig @ 300 degF - MDMT = 10 degF @ 150 psig	40-PI-9005 Redline	EX-101	Light Naphtha to Feed Preheater
			EX-102	Heavy Naphtha to Feed Preheater
			EX-103	Kero to Feed Preheater
	AF-108: - MAWP = 150 psig @ 350 degF - MDMT = -20 degF @ 150 psig	40-PI-9009	PM-101A/B	Naphtha Stabilizer Overhead Pumps
			PV-101	Naphtha Stabilizer Accumulator
	PV-107: - MAWP = FV/100 psig @ 250 degF - MDMT = 10 degF @ 100 psig	40-PI-9009 Redline	PM-101A/B	Naphtha Stabilizer Overhead Pumps
			PV-101	Naphtha Stabilizer Accumulator
	HTR-101: - MAWP = 100 psig @ 300 degF - MDMT = -20 degF @ 100 psig	40-PI-9013	N/A	F-101 Burner Controls
	EX-111: -MAWP Shell = 150 psig @ 350 degF - MDMT Shell = -20 degF @ 150 psig	40-PI-9018	PM-103A/B	COMBI Tower Overhead Pumps
			PV-102	COMBI Tower Accumulator
	PV-109: - MAWP = FV/125 psig @ 250 degF - MDMT = -10 degF @ 100 psig	40-PI-9030 Redline	N/A	Recycle Gas Compressor
		40-PI-9030	N/A	Recycle Gas Compressor
		E000318-01-101 SHT 1 of 5	CS-101A	Compressor Containment Seal Pot
			CM-101A-PM	Seal System Circulating Pump
			CM-101A-BM	Seal System Blower
			CM-101A-AF	Seal System Heat Coil
			CM-101A	Recycle Compressor
		E000318-01-101 SHT 2 of 5	CS-101B	Compressor Containment Seal Pot
			CM-101B-PM	Seal System Circulating Pump

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
			CM-101B-BM	Seal System Blower
			CM-101B	Recycle Compressor
		E000318-01-101 SHT 3 of 5	FL-105A/105B	Compressor Lube Oil Filter
			HTR-102	Lube Oil Separator Heater
			PV-108	Lube Oil Separator Vessel
		E000318-01-101 SHT 4 of 5	CP-101	Control Panel
			CMM-101A/B	Compressor Motor
			CPL-101A/B	Coupling
		E000318-01-101 SHT 5 of 5	AF-108	Gas Aftercooler
			AF-107	Compressor Lube Oil Cooler
		E000318-02-102 SHT 2	PV-109	Discharge Scrubber
			PM-117A/B	Discharge Pump
		E000318-02-102-1	EX-111	Gas to Gas Heat Exchanger
			HTR-101	Electric Gas Preheater
			PM-113A	Scrubber Pump
			PV-107	Suction Scrubber
10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air	Typically run 3 out of 4 hot oil circulation pumps. Hot oil temperature - 600F when exchangers are clean, up to 640-650F when exchangers are fouled. Hot oil to Heat Exchanger EX-106, Heat Exchanger EX-107A/B, Heat Exchanger EX-109, Heat Exchanger EX-110 EX-106: - Shell MAWP = 287 psig @ 650 degF - Shell MDMT = 287 psig @ 20 degF - Tube MAWP = 276 psig @ 700 degF - Tube MDMT = 276 psig @ 20 degF - Duty = 54.3 MMBTU/Hr EX-107: - Shell MAWP = 227 psig @ 650 degF - Shell MDMT = 227 psig @ 20 degF - Tube MAWP = 218 psig @ 700 degF - Tube MDMT = 218 psig @ 20 degF - Duty = 121.7 MMBTU/Hr PV-103: - Design = 50 psig MAWP @ 650 degF PM-102A/B/C/D: - MAWP = 529 psig @ 625 degF	40-PI-9010	EX-106	Naphtha Stabilizer Reboiler
			EX-107	COMBI Tower Feed Preheater
		40-PI-9010 Redline	EX-106	Naphtha Stabilizer Reboiler
			EX-107	COMBI Tower Feed Preheater
		40-PI-9011	PV-103	Hot Oil Surge Tank
			PM-102A/B/C/D	Hot Oil Circulation Pumps
		40-PI-9011 Redline 1	PV-103	Hot Oil Surge Tank
			PM-102A/B/C/D	Hot Oil Circulation Pumps
		40-PI-9011 Redline 2	PV-103	Hot Oil Surge Tank
			PM-102A/B/C/D	Hot Oil Circulation Pumps
		40-PI-9012	F-101	Hot Oil Heater
		40-PI-9019	T-103	Heavy Naphtha Stripper
			EX-108	Heavy Naphtha Stripper Reboiler
		40-PI-9020	T-104	Kero Stripper
			EX-109	Kero Stripper Reboiler
		40-PI-9049	T-105	Diesel Stripper
			EX-110	Diesel Stripper Reboiler
		13040-D-100-1	F-201	Hot Oil Heater

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	- Design = 2433 GPM, 400 HP, 1750 RPM, 536 degF @ 160 psi deltaP			
	EX-108: - Shell MAWP = 83 psig @ 650 degF - Shell MDMT = -20 degF @ 83 psig - Tube MAWP = 156 psig @ 700 degF - Tube MDMT = -20 degF @ 156 psig - Duty = 13.4 MMBTU/Hr			
	EX-109: - Shell MAWP = 258 psig @ 650 degF - Shell MDMT = -20 degF @ 258 psig - Tube MAWP = 298 psig @ 700 degF - Tube MDMT = -20 degF @ 298 psig - Duty = 24.7 MMBTU/Hr			
	EX-110: - Shell MAWP = 96 psig @ 650 degF - Shell MDMT = -20 degF @ 96 psig - Tube MAWP = 186 psig @ 700 degF - Tube MDMT = -20 degF @ 186 psig - Duty = 7.04 MMBTU/Hr			
11. Node 11: Butane/Y Grade Storage Sphere, pumps and truck rack	PV-410 and PV-411 - 20,000 bbl each; pump PM-416A/B/C - 1000 GPM, LR-501/502 loading racks, Water draw off - 2x/shift; Typically only run PM-416C, as this is the only PM-416 pump that has an inline strainer to prevent solids.	40-PI-9038	PV-410	Y-Grade Storage Sphere
		40-PI-9039	PM-416A/B/C	Multi-Stage Vertical Turbine Pump
		40-PI-9046	F-501/502	Truck Unloading Filter
	PV-410: - Design = 120 psig MAWP @ 210/-20 degF and Full Vac @ -20 degF - Capacity = 23,044 bbls		LR-501	Truck Loading Skid
			PM-501	Prover Drain Pump
		40-PI-9051	LR-502	Truck Loading and Unloading Skid
	PM-416A/B/C: - Design = 1000 GPM, 60 HP @ 1785 RPM - Temp = 120 degF		F-503/504	Truck Loading Filter
			PM-502	Prover Drain PO
		40-PI-9052	LR-502	Truck Loading and Unloading Skid
	F-501/502: - Design = 285 psig @ 150 degF - MDMT = -20 degF @ 285 psig		F-505/506	Truck Unloading Filter
		40-PI-9054	PV-411	Butane/Y-Grade Storage Sphere
	LR-501: - Size = 6" - 500 GPM each bay, 1000 GPM total flow - Design = 285 psig @ 100 degF			
	PM-501: - Capacity = 2 GPM - Design = 1 HP @ 1725 RPM			
	LR-502: - Loading Size = 6" - 500 GPM each bay, 1000 GPM total flow - Design = 150 psig @ 100 degF			

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	- Unloading Size = 4" - 160 GPM each bay, 320 GPM total flow - Design = 285 psig @ 100 degF F-503/504: - Design = 285 psig @ 150 degF - MDMT = -20 degF PM-502: - Capacity = 2 GPM - Design = 1 HP @ 172 RPM F-505/506: - Design = 250 psig @ 150 degF - MDMT = -20 degF @ 250 psig PV-411: - Design = 120 psig MAWP @ 210/-20 degF and Full Vac @ -20 degF - Capacity = 23,044 bbls			
12. Node 12: Flare, Flare Knockout Drum, Flare Knockout Drum Pump (PV-305, PM-312) and Flare, Flow switch FL-301	Knockout drum pumps in manual, typically run once/week. Run one flare flower normally, run two blowers if high flow rate is anticipated PV-305: - MAWP = 50 psig @ 300 degF - MDMT = 10 degF PM-312A/B: - Design = 185 GPM, 15 HP, 1750 RPM, 31-119 degF @ deltaP 50 psi	40-PI-9027	PV-305	Flare Knockout Drum
			PM-312A/B	Flare Knockout Pump
		40-PI-9043	N/A	Flare Header
		40-PI-9231	N/A	Flare Header
		YA-1770-S1	N/A	Flare System 3-Pilot
13. Node 13: Aqueous Ammonia Heater System (PV-106, PM-119A/B, BM-105A/B)	Ammonia injection into hot oil heater stack. NH3 vaporized with warmed air. 10,000 gallon storage tank of 19% aqueous NH3, filled ~ quarterly.	40-PI-9012	F-101	Hot Oil Heater
		40-PI-9013	N/A	F-101 Burner Controls
		13J0010_100101	PV-106	Ammonia Storage Tank
			PM-119A/B	NH3 Forwarding Pump
			BM-105A/B	Dilution Air Blower
		13J0039_100101	PM-120A/B	NH3 Forwarding Pump
			BM-205A/B	Dilution Air Blower
14. Node 14: Natural Gas/Fuel Gas Header	Natural gas received from KM pipeline, not odorized	40-PI-9009	PM-101A/B	Naphtha Stabilizer Overhead Pumps
			PV-101	Naphtha Stabilizer Accumulator
		40-PI-9009 Redline	PM-101A/B	Naphtha Stabilizer Overhead Pumps
			PV-101	Naphtha Stabilizer Accumulator
		40-PI-9014	T-102	COMBI Tower
		40-PI-9030 Redline	N/A	Recycle Gas Compressor
		40-PI-9030	N/A	Recycle Gas Compressor
		40-PI-9043	N/A	Flare Header
		40-PI-9048	N/A	Natural Gas Header

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
		40-PI-9235	FL-208	Vertical Filter Separator
		88-P1-9244	N/A	Fuel Gas
15. Node 15: Hydrocarbon Drain	8000 gallon hydrocarbon sump, pumped out to rerun line TK-101: - Design = 5 psig @ 400 degF - Nominal Capacity = 8000 gal PM-118A/B: - Design = 250 GPM, 15 HP, 3500 RPM - Temp = 75 degF @ 35 psi deltaP	40-PI-9044	TK-101	Hydrocarbon Sump
			PM-118A/B	Hydrocarbon Sump Pumps
16. Node 16: Wastewater Drain	8000 gallon sump, pumped to TK 5 TK-102: - Design = 5 psig @ 400 degF - Nominal Capacity = 8000 gal PM-115A/B: - Design = 100 GPM, 5 HP, 3515/3530 RPM - Temp = AMB @ 26 psi deltaP	40-PI-9004 Redline	PM-111A/B/C	Condensate Feed Pumps
			FL-101A/B	Condensate Feed Filter/Coalescer
		40-PI-9004	PM-111A/B/C	Condensate Feed Pumps
			FL-101A/B	Condensate Feed Filter/Coalescer
		40-PI-9009	PM-101A/B	Naphtha Stabilizer Overhead Pumps
			PV-101	Naphtha Stabilizer Accumulator
		40-PI-9009 Redline	PM-101A/B	Naphtha Stabilizer Overhead Pumps
			PV-101	Naphtha Stabilizer Accumulator
		40-PI-9045	TK-102	Contact Water Sump
			PM-115A/B	Contact Water Sump Pumps
17. Node 17: Air Compressor System	Splitter 1 air system and common air header for both splitters; Two compressors for each splitter train, tied into common header. Splitter 2 compressors are larger than Splitter 1. CM-102A/B: - Design = 192 CFM @ 174 psig, 60 HP PV-211: - Design = 200 psig ASME @ 400 degF, 1040 Gal DR-101: - Design = 232 psig @ 100 degF FL-104A/B: - Design = 102 psig @ 68 degF; 254 CFM @ 102 psi FL-106: - Design = 102 psig @ 68 degF; 254 CFM @ 102 psi	40-PI-9041	N/A	Air Compressor/Dryer
		40-PI-9047	N/A	Plant Air/Instrument Air Diagram
		1318317118 SHT 1	CM-102A/B	Air Compressors A, B
		1318317118 SHT 2	PV-111	Air Receiver (Wet)
		1318317118 SHT 3	DR-101	Air Dryer Desiccant
			FL-104 A/B	Service Air Filter
			FL-106	Service Air Filter
		1318317118 SHT 4	FL-107	Service Air Filter

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	FL-107: - Design = 232 psig @ 68 degF; 254 CFM @ 102 psi - MDMT = -20 degF			
18. Node 18: Rerun Manifold Header	Rerun line runs to feed tanks or Tank 100-14. Normally rerun to 100-14 and then move rerun from 100-14 back to feed tanks.	40-PI-9022	N/A	Light Naphtha Product to Storage
19. Node 19: Condensate feed from Condensate feed pumps to Condensate Feed Filter/Coalescer through preheaters to Naphtha Stabilizer Column (Pumps PM-111 A/B/C, filter FL-201 A/B, Heat Exchanger EX-201,202,203, 204, 205)	Feed from 200-1/2/3 at ambient temperature, desalter runs ~200F, desalter pressure ~140 psi, 1475 GPM maximum rate (to be confirmed). Tie in to desalter between downstream of Heat Exchanger EX-204 and upstream of Heat Exchanger EX-205. Coalescers are currently bypassed FL-201A/B: - MAWP = 325 psig @ 120 degF - Capacity = 1460 GPM EX-201: - Shell/Tube MAWP = 305 psig @ 450 degF - Duty = 1.34 MMBTU/Hr EX-202: - Shell/Tube MAWP = 305 psig @ 450 degF - Duty = 13.6 MMBTU/Hr EX-203: - Shell MAWP = 305 psig @ 460 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 33.8 MMBTU/Hr EX-204: - Shell/Tube MAWP = 305 psig @ 650 degF - Duty = 18.9 MMBTU/Hr EX-205: - Shell MAWP = 305 psig @ 650 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 29.7 MMBTU/Hr	40-PI-9204 Redline	FL-201A/B	Condensate Feed Filter/Coalescer
		40-PI-9204	FL-201A/B	Condensate Feed Filter/Coalescer
		40-PI-9205	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9205 Redline	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9206	EX-204	Diesel to Feed Preheater
			EX-205	Bottoms to Feed Preheater
		40-PI-9207	T-201	Naphtha Stabilizer Column
20. Node 20: Naphtha Stabilizer Column, Naphtha Stabilizer Reboiler, Combi Tower Feed Preheater to Combi Tower (T-201, Heat Exchanger EX 206, 207A/B)	Bottoms through reboiler, heated with Therminol, Therminol flow controlled by column temperature. Bottoms feed to Combi Tower heated by Therminol (Heat Exchanger EX-207A/B), Therminol flow controlled by Combi Tower feed temperature. T-201: - Design = 120 psig MAWP @ 375 degF - MDMT = 10 degF @ 120 psig	40-PI-9207	T-201	Naphtha Stabilizer Column
		40-PI-9210	EX-206	Naphtha Stabilizer Reboiler
			EX-207	COMBI Tower Feed Preheater
		40-PI-9210 Redline	EX-206	Naphtha Stabilizer Reboiler
			EX-207	COMBI Tower Feed Preheater
		40-PI-9214	T-202	COMBI Tower

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	EX-206: - Shell MAWP = 287 psig @ 650 degF - Tube MAWP = 276 psig @ 700 degF - Duty = 54.3 MMBTU/Hr			
	EX-207: - Shell MAWP = 227 psig @ 650 degF - Tube MAWP = 218 psig @ 700 degF			
21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)	Stabilizer overhead runs at 90-120F, 90 psig, accumulator vents to recycle compressor system	40-PI-9207	T-201	Naphtha Stabilizer Column
		40-PI-9208	AF-201	Naphtha Stabilizer Condenser
	AF-201: - Design = 120 psig MAWP @ 300 degF - Design Duty = 10.7 MMBTU/Hr	40-PI-9209 Redline	PM-201A/B	Naphtha Stabilizer Overhead Pumps
			PV-201	Naphtha Stabilizer Accumulator
	PV-201: - Design = 120 psig MAWP @ 300 degF - MDMT = 10 degF @ 120 psig	40-PI-9209	PM-201A/B	Naphtha Stabilizer Overhead Pumps
			PV-201	Naphtha Stabilizer Accumulator
	PM-201A/B: - Design = 239 GPM, 15 HP, 3550 RPM - Temp = 120 degF @ 56 psi deltaP	40-PI-9222	N/A	Light Naphtha Product to Storage
		40-PI-9222 Redline	N/A	Light Naphtha Product to Storage
22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)		40-PI-9038	PV-410	Y-Grade Storage Sphere
	Combi Tower Bottoms - 525-560F, 9-11 psig, temperature to storage = 140-160F, product flow rate = 250-450 GPM; bottoms product temperature controlled by turning fin fans on/off	40-PI-9059A	EX-212A/B	Bottoms Product to Desalter Wash Water Heater
			EX-112A/B	Bottoms Product to Desalter Wash Water Heater
	EX-205: - Shell MAWP = 305 psig @ 650 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 29.7 MMBTU/Hr	40-PI-9206	EX-204	Diesel to Feed Preheater
			EX-205	Bottoms to Feed Preheater
		40-PI-9207	T-201	Naphtha Stabilizer Column
	T-202: - Design = 50 psig MAWP @ 575 degF - MDMT = -20 degF @ 50 psig	40-PI-9214	T-202	COMBI Tower
		40-PI-9222	N/A	Light Naphtha Product to Storage
		40-PI-9222 Redline	N/A	Light Naphtha Product to Storage
	PM-204A/B: - Design = 631 GPM, 75 HP, 3550 RPM - Temp = 548 degF @ 105 psi deltaP	40-PI-9225	PM-204A/B	COMBI Tower Bottoms Product Pumps
	AF-203: - Design = 250 psig MAWP @ 625 degF - Design Duty = 49.5 MMBTU/Hr	40-PI-9226	AF-203	Bottoms Product Cooler
	EX-212A/B: - Shell MAWP = 305 psig @ 650 degF			

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	- Tube MAWP = 305 psig @ 290 degF - Duty = 2.3 MMBTU/Hr			
23. Node 23: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-202, 1/12, PV-202, PM-203 A/B, Heat Exchanger EX-201) to Tanks 24/25	Overhead pressure - 9-11 psi, overhead temperature ~188F, temperature to storage - 67-72F after Aggreko chillers. 12 fin fan banks, 24 fans. Typically run all fans, may be able to shut 5 fans in winter. EX-201: - Shell/Tube MAWP = 305 psig @ 450 degF - Duty = 1.34 MMBTU/Hr AF-202: - Design = 50 psig MAWP @ 300 degF - Design Duty = 150.5 MMBTU/Hr PV-202: - Design = 50 psig MAWP @ 300 degF PM-203A/B: - Design = 2528 GPM, 300 HP, 1760 RPM - MAWP = 533 psig @ 139 degF @ 119 psi deltaP	40-PI-9205	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9205 Redline	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9214	T-202	COMBI Tower
		40-PI-9215	AF-202	COMBI Tower Condenser
		40-PI-9216	AF-202	COMBI Tower Condenser
		40-PI-9217	AF-202	COMBI Tower Condenser
		40-PI-9218 Redline	PM-203A/B	COMBI Tower Overhead Pumps
			PV-202	COMBI Tower Accumulator
		40-PI-9218	PM-203A/B	COMBI Tower Overhead Pumps
			PV-202	COMBI Tower Accumulator
		40-PI-9222	N/A	Light Naphtha Product to Storage
		40-PI-9222 Redline	N/A	Light Naphtha Product to Storage
24. Node 24: Heavy Naphtha Side Draw From T-202 to Heavy Naphtha Stripper (T-203), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-202 (T-203, Heat Exchanger EX-208, PM-205A/B, AF-205)	Normally run with no hot oil on reboiler (Heat Exchanger EX-208), cooled to ~90F in AF-205, stripper pressure approximately same as Combi Tower EX-202: - Shell/Tube MAWP = 305 psig @ 450 degF - Duty = 13.6 MMBTU/Hr T-203: - Design 50 psig MAWP @ 400 degF - MAWP = 50 psig @ 10 degF EX-208: - Shell MAWP = 83 psig @ 650 degF - Shell MDMT = -20 degF @ 83 psig - Tube MAWP = 156 psig @ 700 degF - Tube MDMT = -20 degF @ 156 psig - Duty = 13.4 MMBTU/Hr PM-205A/B:	40-PI-9205	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9205 Redline	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9214	T-202	COMBI Tower
		40-PI-9219	T-203	Heavy Naphtha Stripper
			EX-208	Heavy Naphtha Stripper Reboiler
		40-PI-9221 Redline	PM-205A/B	Heavy Naphtha Stripper Product Pumps
			PM-206A/B	Keto Stripper Product Pumps

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	- Design = 493 GPM, 60 HP, 3550 RPM - MAWP = 642 psig @ 350 degF @ 92 psi deltaP AF-205: - Design = 250 psig MAWP @ 375 degF - Design Duty = 15.3 MMBTU/Hr	40-PI-9221	PM-207A/B	Diesel Stripper Product Pumps
			PM-205A/B	Heavy Naphtha Stripper Product Pumps
			PM-206A/B	Keto Stripper Product Pumps
		40-PI-9222	PM-207A/B	Diesel Stripper Product Pumps
			N/A	Light Naphtha Product to Storage
			N/A	Light Naphtha Product to Storage
		40-PI-9223	AF-205	Heavy Naphtha Product Cooler
25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)	Kero stripper runs at ~360-390F, stripper pressure approx. same as Combi Tower EX-203: - Shell MAWP = 305 psig @ 460 degF - Tube MAWP = 305 psig @ 450 degF - Duty = 33.8 MMBTU/Hr T-204: - Design = 50 psig MAWP @ 600 degF - MDMT = 10 degF @ 50 psig EX-209: - Shell MAWP = 258 psig @ 650 degF - Shell MDMT = -20 degF @ 258 psig - Tube MAWP = 298 psig @ 700 degF - Tube MDMT = -20 degF @ 298 psig - Duty = 24.7 MMBTU/Hr PM-206A/B: - Design = 978 GPM, 100 HP, 1750 RPM - Temp = 412 degF @ 85 psi deltaP AF-204: - Design = 250 psig MAWP @ 450 degF - Design Duty = 46.4 MMBTU/Hr	40-PI-9205	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9205 Redline	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9214	T-202	COMBI Tower
		40-PI-9220	T-204	Kero Stripper
			EX-209	Kero Stripper Reboiler
		40-PI-9221	PM-205A/B	Heavy Naphtha Stripper Product Pumps
			PM-206A/B	Keto Stripper Product Pumps
			PM-207A/B	Diesel Stripper Product Pumps
		40-PI-9221 Redline	PM-205A/B	Heavy Naphtha Stripper Product Pumps
			PM-206A/B	Keto Stripper Product Pumps
			PM-207A/B	Diesel Stripper Product Pumps
		40-PI-9222	N/A	Light Naphtha Product to Storage
		40-PI-9222 Redline	N/A	Light Naphtha Product to Storage
		40-PI-9224	AF-204	Kero Product Cooler
26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)	Diesel stripper runs at ~525-530F, stripper pressure approx. same as Combi Tower EX-204:	40-PI-9206	EX-204	Diesel to Feed Preheater
			EX-205	Bottoms to Feed Preheater
		40-PI-9214	T-202	COMBI Tower

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	- Shell/Tube MAWP = 305 psig @ 650 degF - Duty = 18.9 MMBTU/Hr PM-207A/B: - Design = 320 GPM, 50 HP, 3575 RPM - MAWP = 543 psig @ 650 degF @ 106 psi deltaP T-205: - Design = 57 psig MAWP @ 650 degF - MDMT = 10 degF @ 57 psig EX-210: - Shell MAWP/MDMT = 96 psig @ 650 degF/ -20 degF - Tube MAWP/MDMT = 186 psig @ 700 degF/ -20 degF - Duty = 7.04 MMBTU/Hr AF-206: - MAWP = 250 psig @ 700 degF - Design Duty = 26.14 MMBTU/Hr	40-PI-9221 Redline	PM-205A/B	Heavy Naphtha Stripper Product Pumps
			PM-206A/B	Keto Stripper Product Pumps
			PM-207A/B	Diesel Stripper Product Pumps
		40-PI-9221	PM-205A/B	Heavy Naphtha Stripper Product Pumps
			PM-206A/B	Keto Stripper Product Pumps
			PM-207A/B	Diesel Stripper Product Pumps
		40-PI-9222	N/A	Light Naphtha Product to Storage
		40-PI-9222 Redline	N/A	Light Naphtha Product to Storage
		40-PI-9236	T-205	Diesel Stripper
			EX-210	Diesel Stripper Reboiler
		40-PI-9237	AF-206	Diesel Product Cooler
27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)	Inlet pressure 8-12 psig to ~70 psig through compressor, liquids knocked out and returned to Stabilizer feed CM-201A/B: - Suction P = 16 psia @ 165 degF - Discharge P = 80 psia @ 265 degF PV-208: - MAWP = 150 psig @ 300 degF - MDMT = 10 degF @ 150 psig AF-208: - MAWP = 150 psig @ 350 degF - MDMT = -20 degF @ 150 psig PV-207: - MAWP = FV/100 psig @ 250 degF - MDMT = 10 degF @ 100 psig HTR-201: - MAWP = 100 psig @ 300 degF - MDMT = -20 degF @ 100 psig EX-211: - MAWP Shell = 150 psig @ 250 degF - MDMT Shell = -20 degF @ 150 psig - MAWP Tube = 150 psig @ 350 degF - MDMT Tube = -20 degF @ 150 psig PV-209: - MAWP = FV/125 psig @ 250 degF - MDMT = -10 degF @ 100 psig	40-PI-9205	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9205 Redline	EX-201	Light Naphtha to Feed Preheater
			EX-202	Heavy Naphtha to Feed Preheater
			EX-203	Kero to Feed Preheater
		40-PI-9209 Redline	PM-201A/B	Naphtha Stabilizer Overhead Pumps
			PV-201	Naphtha Stabilizer Accumulator
		40-PI-9209	PM-201A/B	Naphtha Stabilizer Overhead Pumps
			PV-201	Naphtha Stabilizer Accumulator
		40-PI-9213	N/A	F-201 Burner Controls
		40-PI-9218 Redline	PM-203A/B	COMBI Tower Overhead Pumps
			PV-202	COMBI Tower Accumulator
		40-PI-9218	PM-203A/B	COMBI Tower Overhead Pumps
			PV-202	COMBI Tower Accumulator
		40-PI-9228 Redline	N/A	Recycle Gas Compressor
		40-PI-9228	N/A	Recycle Gas Compressor

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
		E000353-01-101 SHT 1	CS-201A	Compressor Containment Seal Pot
			CM-201A-PM	Seal System Circulating Pump
			CM-201A	Recycle Compressor
			CM-201A-BM	Seal System Blower
			CM-201A-AF	Seal System Heat Coil
		E000353-01-101 SHT 2	CM-201B-AF	Seal System Heat Coil
			CS-201B	Compressor Containment Seal Pot
			CM-201B-PM	Seal System Circulating Pump
			CM-201B-BM	Seal System Blower
			CM-201B	Recycle Compressor
		E000353-01-101 SHT 3	FL-205A/B	Compressor Lube Oil Filter
			HTR-202	Lube Oil Separator Heater
			PV-208	Lube Oil Separator Vessel
		E000353-01-101 SHT 4	CMM-201A/B	Compressor Motor
		E000353-01-101 SHT 5	AF-208	Gas Aftercooler
			AF-207	Compressor Lube Oil Cooler
		E000353-02-102 SHT 1	PV-207	Recycle Compressor Inlet Scrubber
		E000353-02-102 SHT 2	PV-209	Discharge Scrubber
			PM-217A/B	Discharge Pump
28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air	Typically run 3 out of 4 hot oil circulation pumps. Hot oil temperature - 600F when exchangers are clean, up to 640-645F when exchangers are fouled. Hot oil to Heat Exchanger EX-206, Heat Exchanger EX-207A/B, Heat Exchanger EX-209, Heat Exchanger EX-210. Nitrogen blanket on surge tank from N2 bottles, pressure maintained at 25 psig EX-206: - Shell MAWP = 287 psig @ 650 degF - Tube MAWP = 276 psig @ 700 degF - Duty = 54.3 MMBTU/Hr EX-207:	40-PI-9210	EX-206	Naphtha Stabilizer Reboiler
			EX-207	COMBI Tower Feed Preheater
		40-PI-9210 Redline	EX-206	Naphtha Stabilizer Reboiler
			EX-207	COMBI Tower Feed Preheater
		40-PI-9211 Redline 2	PV-203	Hot Oil Surge Tank
			PM-202A/B/C/D	Hot Oil Circulation Pumps
		40-PI-9211 Redline	PV-203	Hot Oil Surge Tank
			PM-202A/B/C/D	Hot Oil Circulation Pumps
		40-PI-9211	PV-203	Hot Oil Surge Tank
			PM-202A/B/C/D	Hot Oil Circulation Pumps
		40-PI-9212	F-201	Hot Oil Heater

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	- Shell MAWP = 227 psig @ 650 degF - Tube MAWP = 218 psig @ 700 degF PV-203: - Design = 50 psig MAWP @ 650 degF PM-202A/B/C: - Design = 2433 GPM, 400 HP, 1750 RPM - Temp = 536 degF @ 160 psi deltaP EX-208: - Shell MAWP = 83 psig @ 650 degF - Shell MDMT = -20 degF @ 83 psig - Tube MAWP = 156 psig @ 700 degF - Tube MDMT = -20 degF @ 156 psig - Duty = 13.4 MMBTU/Hr EX-209: - Shell MAWP = 258 psig @ 650 degF - Shell MDMT = -20 degF @ 258 psig - Tube MAWP = 298 psig @ 700 degF - Tube MDMT = -20 degF @ 298 psig - Duty = 24.7 MMBTU/Hr EX-210: - Shell MAWP/MDMT = 96 psig @ 650 degF/ -20 degF - Tube MAWP/MDMT = 186 psig @ 700 degF/ -20 degF - Duty = 7.04 MMBTU/Hr	40-PI-9219	T-203	Heavy Naphtha Stripper
			EX-208	Heavy Naphtha Stripper Reboiler
		40-PI-9220	T-204	Kero Stripper
			EX-209	Kero Stripper Reboiler
		40-PI-9236	T-205	Diesel Stripper
			EX-210	Diesel Stripper Reboiler
29. Node 29: Aqueous Ammonia Heater System	Ammonia injection into hot oil heater stack. NH3 vaporized with warmed air. 10,000 gallon storage tank of 19% aqueous NH3, filled ~ quarterly PV-106: - Capacity = 10,000 gal PM-119A/B: - Design = 0.5 HP @ 1750 RPM - Max Capacity = 3 GPM - Max discharge P = 100 psig BM-105A/B: - Design = 5 HP @ 3510 RPM - Capacity = 192 CFM @ 42" W.C. PM-120A/B: - Design = 0.5 HP @ 1750 RPM - Max Capacity = 3 GPM - Max Discharge P = 100 psig BM-205A/B: - Design = 5 HP @ 3510 RPM - Capacity = 192 CFM @ 42" W.C.	40-PI-9212	F-201	Hot Oil Heater
		40-PI-9213	N/A	F-201 Burner Controls
		13J0010_100101	PV-106	Ammonia Storage Tank
			PM-119A/B	NH3 Forwarding Pump
			BM-105A/B	Dilution Air Blower
		13J0039_100101	PM-120A/B	NH3 Forwarding Pump
			BM-205A/B	Dilution Air Blower
30. Node 30: Natural Gas/Fuel Gas Header	Natural gas received from KM pipeline, not odorized	40-PI-9228	N/A	Recycle Gas Compressor

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	FL-208: - MAWP = 285 psig @ 100 degF	40-PI-9228 Redline	N/A	Recycle Gas Compressor
		40-PI-9231	N/A	Flare Header
		40-PI-9235	FL-208	Vertical Filter Separator
		88-P1-9244	N/A	Fuel Gas
31. Node 31: Hydrocarbon Drain	8000 gallon hydrocarbon sump, pumped out to rerun line TK-201: - Design = 5 psig @ 400 degF - Nominal Capacity = 8000 gal PM-218A/B: - Design = 250 GPM, 15 HP, 3530/3550 RPM - Temp = 75 degF @ 35 psi deltaP	40-PI-9232	TK-201	Hydrocarbon Sump
			PM-218A/B	Hydrocarbon Sump Pumps
32. Node 32: Wastewater Drain	8000 gallon sump, pumped to TK 5, option to bypass sump directly to storm water pond (first inch goes to TK 5) TK-202: - Design = 5 psig @ 400 degF - Nominal Capacity = 8000 gal PM-215A/B: - Design = 100 GPM, 5 HP, 3500/3530 RPM - Temp = Ambient @ 26 psi deltaP	40-PI-9209 Redline	PM-201A/B	Naphtha Stabilizer Overhead Pumps
			PV-201	Naphtha Stabilizer Accumulator
		40-PI-9209	PM-201A/B	Naphtha Stabilizer Overhead Pumps
			PV-201	Naphtha Stabilizer Accumulator
		40-PI-9232	TK-201	Hydrocarbon Sump
			PM-218A/B	Hydrocarbon Sump Pumps
		40-PI-9233	TK-202	Contact Water Sump
			PM-215A/B	Contact Water Sump Pumps
33. Node 33: Air Compressor System	Two compressors for each splitter train, tied into common header. Splitter 2 compressors are larger than Splitter 1 (375 CFM vs. 192 CFM) CM-202A/B: - Design = 375 CFM @ 174 psig; 125 HP PV-211: - Design 200 psig ASME @ 400 degF; 2200 gal DR-201: - Design = 181 psig @ 100 degF FL-204A/B: - Design = 102 psig @ 68 degF, 360 CFM @ 102 psi FL-206: - Design = 102 psig @ 68 degF, 360 CFM @ 102 psi FL-207: - Design = 102 psig @ 68 degF, 1165 CFM @ 102 psi	1318317146 SHT 1	CM-202A/B	Air Compressors A, B
		1318317146 SHT 2	PV-211	Air Receiver (Wet)
		1318317146 SHT 3	DR-201	Air Dryer Desiccant
			FL-204A/B	Service Air Filter
			FL-206	Service Air Filter
		1318317146 SHT 4	FL-207	Service Air Filter
34. Node 34: Condensate Feed through Desalter	From 2019 Desalter Project PHA.	40-PI-9006	EX-104	Diesel to Feed Preheater

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	Two desalter systems (Splitter 1 and Splitter 2). Flow: Designed for 50,000 bpd per desalter. Actual flow expected to vary between 33,000 to 54,000 bpd. Three Booster Pumps in parallel (shared between the two systems). Normal operation: PM-121A feeds Desalter 1, PM-121B feeds Desalter 2, and PM-121C is spare. Booster pump suction = 83.5psig @ 190 - 210F (210F maximum), discharge = 164psig @ 210F, dP = 40psi. 42API case. Line rated for 600psig (ANSI 300). Demulsifier system pumps are positive displacement. EAD (at least one) present in booster pump area. Feed pumps max discharge = 285psig. Maximum pressure of desalter system = 325psig (dead head of feed pump plus booster pump max increase). Control room manned 24/7. Desalter vessel has 4 level transmitters (condensate and water level). Residence time in vessel is approximately 40 min. PM-121A/B: - Design = 1600 GPM, 60 HP, 1800 RPM - Temp = 210 degF @ 40 psi deltaP D-101/201: - Design = 305 psi @ 650 degF TK-108: - Capacity = 3000 gal - Design = 100 degF; Atm PM-126A/B: - Capacity = 1.8 GPH - Design T = 100 degF	40-PI-9057 40-PI-9058 40-PI-9059A 40-PI-9064 40-PI-9066 40-PI-9206 40-PI-9241	EX-105	Bottoms to Feed Preheater
			PM-121A/B/C	Condensate Booster Pumps
			DT-101A/B/C	Desalter Transformer
			D-101	Condensate Desalter
			EX-212A/B	Bottoms Product to Desalter Wash Water Heater
			EX-112A/B	Bottoms Product to Desalter Wash Water Heater
			N/A	Desalter Sample System
			TK-108	Demulsifier Chemical Storage Tank
			PM-126A/B	Demulsifier Metering Pump
			EX-204	Diesel to Feed Preheater
			EX-205	Bottoms to Feed Preheater
			D-201A/B/C	Desalter Transformer
			D-201	Condensate Desalter
35. Node 35: Wash Water System	From 2019 Desalter Project PHA. TK-103 Treated Wash Water Storage Tank is currently bypassed. Minimum wash water flow = 1500bpd (3% of condensate feed rate). Maximum flow: 2500bpd (5%). Supplied by the city at 70-80psig. Two wash water pumps (operationally set such that PM-122A feeds Splitter 1 and PM-122B feeds Splitter 2). Wash water pumps discharge at 184psig. 144psig upon entering desalter. Water is at ambient temperature (estimated high = 85F). 1hr residence time in storage tank. EX-113A/B/C: - Shell MAWP = 305 psig @ 290 degF	40-PI-9058	DT-101A/B/C	Desalter Transformer
			D-101	Condensate Desalter
		40-PI-9059	EX-113A/B/C	Wash Water to Desalter Waste Water Preheater
		40-PI-9059A	EX-212A/B	Bottoms Product to Desalter Wash Water Heater
			EX-112A/B	Bottoms Product to Desalter Wash Water Heater
		40-PI-9060	PK-001	Water Softening System
			TK-106	Brine Day Tank
			TK-103	Treated Wash Water Storage Tank
			PM-122A/B	Desalter Wash Water Pump

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	- Tube MAWP = 305 psig @ 290 degF - Duty = 2.5 MMBTU/Hr EX-212A/B: - Shell MAWP = 305 psig @ 650 degF - Tube MAWP = 305 psig @ 290 degF - Duty = 2.3 MMBTU/Hr EX-112A/B: - Shell MAWP = 305 psig @ 650 degF - Tube MAWP = 305 psig @ 290 degF - Duty = 2.3 MMBTU/Hr PK-001: - GPM = 150 TK-103: - Capacity = 5250 gal PM-122A/B: - Design = 170 GPM, 50 HP, 3600 RPM - Temp = 85 degF @ 187 psi deltaP TK-107: - Capacity = 4500 gal - Design T = 100 degF - Design P = Atm PM-124A/B: - Design = 15 GPM, 1/3 HP, 3450 RPM PM-127: - Capacity = 0.9 GPH - Discharge P = 126 psig - Design T = 100 degF TK-105: - Design T = 100 degF - Design P = Atm	40-PI-9061	AF-109	Desalter Waste Water Cooler
			AF-209	Desalter Waste Water Cooler
		40-PI-9063	TK-107	Sodium Bi-Sulfite Storage Tank
			PM-124A/B	Brine Transfer Pumps
			PM-127	Sodium Bi-Sulfite Metering Pump
		40-PI-9064	TK-105	Salt/Brine Storage Tank
			N/A	Desalter Sample System
		40-PI-9241	D-201A/B/C	Desalter Transformer
			D-201	Condensate Desalter
36. Node 36: Waste Water	From 2019 Desalter Project PHA. Minimum water flow = 1500bpd (3% of condensate feed rate). Maximum flow: 2500bpd (5%). Steady state. Exits desalter at approximately 144psig. Water exits desalter at 210F (234F max). Water removal controlled by level of desalter. EX-113A/B/C: - Shell MAWP = 305 psig @ 290 degF - Tube MAWP = 305 psig @ 290 degF - Duty = 2.5 MMBTU/Hr AF-109/209: - Design = 305 psig MAWP @ 290 degF - Design Duty = 4 MMBTU/Hr	40-PI-9045	TK-102	Contact Water Sump
			PM-115A/B	Contact Water Sump Pumps
		40-PI-9058	DT-101A/B/C	Desalter Transformer
			D-101	Condensate Desalter
		40-PI-9059	EX-113A/B/C	Wash Water to Desalter Waste Water Preheater
		40-PI-9061	AF-109	Desalter Waste Water Cooler
			AF-209	Desalter Waste Water Cooler
		40-PI-9241	D-201A/B/C	Desalter Transformer
			D-201	Condensate Desalter

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
37. Node 37: Mud Wash	From 2019 Desalter Project PHA. Currently PM-123A is not running due to misalignment issues. Only PM-123B/C running. Occasionally operated to disturb the waste water in the desalters to suspend sediment. 2 pumps in parallel. One pump for each desalter under normal operations (capability to use either pump for either desalter). Flow: maximum = 125gpm. Normal operation is for 70-100gpm. Feed to pumps is 144psig. All valves in line are normally open except the desalter 1 and 2 isolation valves. 30 minutes per wash. Does not interrupt normal operations. Mud wash header in bottom of desalter. Approximate dP = 25psi (169psig total). Planned operation is daily. PM-123A/B/C: - Design = 50 GPM, 10 HP, 3500 RPM - Temp = 210 degF @ 84 psi deltaP	40-PI-9058	DT-101A/B/C	Desalter Transformer
			D-101	Condensate Desalter
		40-PI-9058A	PM-123A/B/C	Desalter Mud Wash Pumps
		40-PI-9241	D-201A/B/C	Desalter Transformer
			D-201	Condensate Desalter
38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25	From 2020 Tank Farm Project PHA This node includes the naphtha rundown chiller from Splitter 1 and/or 2 to Light Virgin Naphtha (LVN) Tanks TK-120-24 or TK-120-25. Also includes recirculation chiller which will cool the naphtha contents of Light Naphtha Tank TK-120-24 or TK-120-25 by recirculation through the chiller by Recirculation Pump P-6. Normal operation may include one or two LVN tanks. Under one LVN tank operation the rundown and recirculation cooler operate on the same tank, under two tank operation, rundown and recirculation may be on the same tank or on opposite tanks. The chillers are mainly used in the summer season. The tanks' target temperature is set based on analysis of the tanks' Reid vapor pressure (RVP) and a calculation of the true vapor pressure (TVP) of the tank. Current operation is to rundown into one of two tanks and recirculate/ discharge from the other tank. Naphtha is transferred from TK-120-24/25 and TK-100-15 via Naphtha Cargo Pumps P-2A/B and 16" pipeline to Galena Park Terminal ground or water shipping area.	88-P1-9217	TK-200-3	Tank - 9000 Area Feedstock
			MIX-9103	Mixer - 9000 Area Feedstock Tank
		88-P1-9226	P-6	Pump - 9000 Area Naphtha
		88-P1-9228	P-2A/B	Pumps - 9000 Area Naphtha
		88-P1-9231	TK-100-15	Tank - 9000 Area Naphtha Heavy
			MIX-9402	Mixer - 9000 Area Naphtha Heavy Tank
		88-P1-9236	MIX-9409	Mixer - 9000 Naphtha Light Tank
			TK-120-24	9000 Naphtha Light Tank
		88-P1-9237	MIX-9401	Mixer - Naphtha Heavy/Light Tank
			TK-120-25	Naphtha Heavy/Light Tank
		88-P1-9239	N/A	Clinton Drive Pipe Interfaces
	From 2020 Tank Farm Project PHA	88-P1-9226	P-6	Pump - 9000 Area Naphtha

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25	This node includes the heavy naphtha rundown from Splitter 1 and/or 2 to Heavy Virgin Naphtha (HVN) Tanks TK-100-15 and TK-120-25 can be used as swing tank. Normal operation may include one or two HVN tanks. Naphtha is transferred from TK-120-25 and TK-100-15 via Naphtha Cargo Pumps P-2A/B and 16" pipeline to Galena Park Terminal ground or water shipping area.	88-P1-9228	P-2A/B	Pumps - 9000 Area Naphtha
		88-P1-9231	TK-100-15	Tank - 9000 Area Naphtha Heavy
			MIX-9402	Mixer - 9000 Area Naphtha Heavy Tank
		88-P1-9237	MIX-9401	Mixer - Naphtha Heavy/Light Tank
			TK-120-25	Naphtha Heavy/Light Tank
		88-P1-9239	N/A	Clinton Drive Pipe Interfaces
40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps	From 2020 Tank Farm Project PHA This node includes the kero rundown from Splitter 1 and/or 2 through the Kero Filters FLT-11/ 12A/B/C/ 13A/B or FLT-21/ 22A/B/C/ 23A/B to Kero Tanks TK-100-20/100-11 and TK-150-10. Normal operation is three kero tanks and rundown through the filters. Kero can be transferred between Kero Tanks or through Kero Filters via Recirculation Pump P-4 or through Kero Transfer Pumps P-1A/B to 16" pipeline to Galena Park Terminal ground or water shipping area. TK-150-10: - Capacity = 150,165 BBI - Design = Atm @ 200 degF - MDMT = 30 degF TK-100-11: - Capacity = 100,000 BBI - Design = Atm @ 200 degF - MDMT = 30 degF	88-P1-9220	TK-150-10	Tank - 9000 Area Kerosene
		88-P1-9221	MIX-9201	Mixer - 9000 Area Kerosene Tank
			TK-100-11	Tank - 9000 Area Kerosene
		88-P1-9222	MIX-9204	Mixer - 9000 Area Kerosene Tank
			TK-100-12	Tank - 9000 Area Kerosene
		88-P1-9229	MIX-9306	Mixer - 9000 Area Kerosene Tank
			P-1A/B	Pumps - 9000 Area Kerosene / Diesel
		88-P1-9232	MIX-9202	Mixer - 9000 Area Diesel Tank
			TK-100-20	Tank - 9000 Area Diesel
		88-P1-9234	P-4	Kerosene/Diesel Cargo Pump
		88-P1-9239	N/A	Clinton Drive Pipe Interfaces
		88-P1-9235	P-KF	Filter Drain Pump
			FLT-11 AND -21	Particulate Filters
			FLT-12A/B/C AND 22A/B/C	"Clay" Filters
			FLT-13A/B AND -23A/B	Coalescing Filters
		88-P1-9217	TK-200-3	Tank - 9000 Area Feedstock
			MIX-9103	Mixer - 9000 Area Feedstock Tank
		88-P1-9224	HTR-9308A/B/C/D	Heaters - 9000 Area AGO/Feedstock
			TK100-13	Tank - 9000 Area AGO/Feedstock
			MIX-9308	Mixer - 9000 Area AGO/Feedstock Tank
		88-P1-9227 Redline	P-7	Pump - 9000 Area Feedstock
			TK-100-14	Tank - 9000 Area Feedstock

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
		88-P1-9227	MIX-9400	Mixer - 9000 Area Feedstock Tank
			P-7	Pump - 9000 Area Feedstock
			TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank
		88-P1-9230	P-5	Pump - 9000 Area AGO
		88-P1-9231	TK-100-15	Tank - 9000 Area Naphtha Heavy
			MIX-9402	Mixer - 9000 Area Naphtha Heavy Tank
41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps	From 2020 Tank Farm Project PHA This node includes the diesel rundown from Splitter 1 and/or 2 to Diesel Tanks TK-100-12/ 21. Normal operation is two diesel tanks. Diesel can be transferred between Diesel Tanks via Recirculation Pump P-4. Diesel Transfer Pumps P-1A/B send diesel to 16" pipeline to Galena Park Terminal ground or water shipping area. Diesel can be used to flush AGO lines and sent to feed pumps. Diesel flush can be returned to diesel tanks. TK-100-12: - Capacity = 100,000 BBI - Design = Atm @ 200 degF - MDMT = 30 degF	88-P1-9216	TK-200-2	Tank - 9000 Area Feedstock
			MIX-9102	Mixer - 9000 Area Feedstock Tank
		88-P1-9217	TK-200-3	Tank - 9000 Area Feedstock
			MIX-9103	Mixer - 9000 Area Feedstock Tank
		88-P1-9218	HX1 & 2	9000 Area Heat Exchangers
		88-P1-9219	MIX-9305	Mixer - AGO
			TK-120-22	Tank - AGO
			HTR-9305A/B/C/D	Heaters - AGO
		88-P1-9221	TK-100-11	Tank - 9000 Area Kerosene
			MIX-9204	Mixer - 9000 Area Kerosene Tank
		88-P1-9222	TK-100-12	Tank - 9000 Area Kerosene
			MIX-9306	Mixer - 9000 Area Kerosene Tank
		88-P1-9223	MIX-9307	Mixer - 9000 Area AGO Tank
			TK-120-23	Tank - 9000 Area AGO
			HTR-9307A/B/C/D	Heaters - 9000 Area AGO
		88-P1-9224	HTR-9308A/B/C/D	Heaters - 9000 Area AGO/Feedstock
			TK100-13	Tank - 9000 Area AGO/Feedstock
			MIX-9308	Mixer - 9000 Area AGO/Feedstock Tank
		88-P1-9225	P-3A/B	Pumps - 9000 Area AGO
		88-P1-9226	P-6	Pump - 9000 Area Naphtha
		88-P1-9227	P-7	Pump - 9000 Area Feedstock

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
			TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank
		88-P1-9227 Redline	P-7	Pump - 9000 Area Feedstock
			TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank
		88-P1-9229	P-1A/B	Pumps - 9000 Area Kerosene / Diesel
		88-P1-9230	P-5	Pump - 9000 Area AGO
		88-P1-9231	TK-100-15	Tank - 9000 Area Naphtha Heavy
			MIX-9402	Mixer - 9000 Area Naphtha Heavy Tank
		88-P1-9233	MIX-9203	Mixer - 9000 Area Diesel Tank
			TK100-21	Tank - 9000 Area Diesel
		88-P1-9234	P-4	Kerosene/Diesel Cargo Pump
		88-P1-9239	N/A	Clinton Drive Pipe Interfaces
		88-P1-9246	PM-111A/B/C	Condensate Feed Pumps
			P-HX	Condensate Feed Pump
42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps	From 2020 Tank Farm Project PHA This node includes the AGO rundown from Splitter 1 and/or 2 to AGO Tanks TK-100-13/ 120-22/23. Normal operation is three AGO tanks. One tank of the three tanks will be set as low, medium and high quality. AGO can be transferred between AGO Tanks via Recirculation Pump P-5. AGO Transfer Pumps P-3A/B send AGO to 16" pipeline to Galena Park Terminal ground or water shipping area, or return to a feed tank. The AGO pig launcher was intended to pig the AGO transfer lines to the terminal. It has never been used in favor of diesel flushing the AGO lines. Heat Exchangers HX1/2 Exchangers can be used to circulate the feed tanks through HX1/2 and pre-heat the feed to the splitters as an energy efficiency. This is not	88-P1-9217	TK-200-3	Tank - 9000 Area Feedstock
			MIX-9103	Mixer - 9000 Area Feedstock Tank
		88-P1-9218	HX1 & 2	9000 Area Heat Exchangers
		88-P1-9219	MIX-9305	Mixer - AGO
			TK-120-22	Tank - AGO
			HTR-9305A/B/C/D	Heaters - AGO
		88-P1-9223	MIX-9307	Mixer - 9000 Area AGO Tank
			TK-120-23	Tank - 9000 Area AGO
			HTR-9307A/B/C/D	Heaters - 9000 Area AGO
		88-P1-9224	HTR-9308A/B/C/D	Heaters - 9000 Area AGO/Feedstock
			TK100-13	Tank - 9000 Area AGO/Feedstock
			MIX-9308	Mixer - 9000 Area AGO/Feedstock Tank
		88-P1-9225	P-3A/B	Pumps - 9000 Area AGO

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	used in practice due to high solids and pump plugging as well as potential environmental concern with heating the feed tanks. HX-1/2: - Shell MAWP = 150 psig @ 650 degF - Tube MAWP = 150 psig @ 650 degF - MDMT = - 20 degF TK-120-22: - Design = Atm @ 200 degF - MDMT = 30 degF TK-120-23: - Capacity = 120,000 BBL - Design = Atm @ 200 degF - MDMT = 30 degF TK100-13: - Capacity = 100,000 BBL	88-P1-9226	P-6	Pump - 9000 Area Naphtha
		88-P1-9227	P-7	Pump - 9000 Area Feedstock
		TK-100-14	TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank
		88-P1-9227 Redline	P-7	Pump - 9000 Area Feedstock
			TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank
		88-P1-9230	P-5	Pump - 9000 Area AGO
		88-P1-9231	TK-100-15	Tank - 9000 Area Naphtha Heavy
			MIX-9402	Mixer - 9000 Area Naphtha Heavy Tank
		88-P1-9239	N/A	Clinton Drive Pipe Interfaces
43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C	From 2020 Tank Farm Project PHA This node includes Feed Tanks TK-200-1/2/3. Normal operation is three Feed tanks with no quality difference. The plant can be fed from 12" Magellan line to all three tanks and TK-100-14 and a 20" Kinder Morgan Condensate Crude KMCC line to all three tanks and TK-100-14. The Magellan line also includes a pig launcher that is operated by Galena Park. The pig launcher on the KMCC line is not operated by Galena Park. Feed can be transferred between Feed Tanks via Pump P-HX which send feed through the AGO Exchanger HX1/2. Feed Pumps PM-111A/B/C send Feed to Splitter 1/2 from TK-200-1/2/3. Feed can be transferred from TK-100-14 to TK-200-1/2/3 via P-7 and then through P-HX also through the AGO Exchanger HX1/2 which will be review in node 7. This node also includes the hydrocarbon header which is a maintenance drain from the splitter, removes liquid from flare knockout drum and is the relief header for the Feed Pumps PM-111A/B/C thermal safeties. TK-200-1/200-3: - Capacity = 200,130 Atm BBL - MDMT = 30 degF - Max Temp = 200 degF	88-P1-9215	TK-200-1	Tank - 9000 Area Feedstock
			MIX-9101	Mixer - 9000 Area Feedstock Tank
		88-P1-9216	TK-200-2	Tank - 9000 Area Feedstock
			MIX-9102	Mixer - 9000 Area Feedstock Tank
		88-P1-9217	TK-200-3	Tank - 9000 Area Feedstock
			MIX-9103	Mixer - 9000 Area Feedstock Tank
		88-P1-9218	HX1 & 2	9000 Area Heat Exchangers
		88-P1-9224	HTR-9308A/B/C/D	Heaters - 9000 Area AGO/Feedstock
			TK100-13	Tank - 9000 Area AGO/Feedstock
			MIX-9308	Mixer - 9000 Area AGO/Feedstock Tank
		88-P1-9227 Redline	P-7	Pump - 9000 Area Feedstock
			TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank
		88-P1-9227	P-7	Pump - 9000 Area Feedstock
			TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	TK-200-2: - Capacity = 200,130 BBI - MAWP = Atm	88-P1-9239	N/A	Clinton Drive Pipe Interfaces
		88-P1-9246	PM-111A/B/C	Condensate Feed Pumps
			P-HX	Condensate Feed Pump
		88-P1-9251	N/A	W. Columbia P/L Connection to Splitter
44. Node 44: Rerun Tank TK-100-14 and Rerun Pump P-HX	From 2020 Tank Farm Project PHA This node includes Rerun Tank TK-100-14. TK-100-14 can be used as a feed surge tank but is normally used in Rerun service which takes finished products after coolers. TK-100-14 can also be fed by the Magellan and KMCC pipelines which was covered in Node 6. Feed or Rerun can be transferred from TK-100-14 to Feed Tanks TK-200-1/2/3 via Pump P-7 and then through P-HX also through the AGO Exchanger HX1/2 to each tank. HX-1/2: - Shell MAWP = 150 psig @ 650 degF - Tube MAWP = 150 psig @ 650 degF - MDMT = - 20 degF	88-P1-9215	TK-200-1	Tank - 9000 Area Feedstock
			MIX-9101	Mixer - 9000 Area Feedstock Tank
		88-P1-9216	TK-200-2	Tank - 9000 Area Feedstock
			MIX-9102	Mixer - 9000 Area Feedstock Tank
		88-P1-9217	TK-200-3	Tank - 9000 Area Feedstock
			MIX-9103	Mixer - 9000 Area Feedstock Tank
		88-P1-9218	HX1 & 2	9000 Area Heat Exchangers
		88-P1-9224	HTR-9308A/B/C/D	Heaters - 9000 Area AGO/Feedstock
			TK100-13	Tank - 9000 Area AGO/Feedstock
			MIX-9308	Mixer - 9000 Area AGO/Feedstock Tank
		88-P1-9227 Redline	P-7	Pump - 9000 Area Feedstock
			TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank
		88-P1-9227	P-7	Pump - 9000 Area Feedstock
			TK-100-14	Tank - 9000 Area Feedstock
			MIX-9400	Mixer - 9000 Area Feedstock Tank
		88-P1-9246	PM-111A/B/C	Condensate Feed Pumps
			P-HX	Condensate Feed Pump
45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19	From 2020 Tank Farm Project PHA This node includes Waste Water Tank TK-5. TK-5 receives process water from Splitter 1/2 Overhead Accumulator PV-102/202 as well as the curbed containment areas. TK-5 also receives water from the two lift stations in the tank farm area which receive water draw from each of the product tanks and the curbed	88-P1-9215	TK-200-1	Tank - 9000 Area Feedstock
			MIX-9101	Mixer - 9000 Area Feedstock Tank
		88-P1-9238	P13 thru P16	Pumps - Process Waste Water Tank
			TK 5-0	Process Waste Water Tank
		88-P1-9238 Redline	P13 thru P16	Pumps - Process Waste Water Tank

Nodes	Design Conditions/Parameters	Drawings / References	Equipment	
			Tag	Description
	containment areas. The water draw from the spheres is isolated. The lift stations send water to TK-5 using Pumps P-13/14/15/16. Pump P-17 and PM-18A/B send water to gulf coast water authority. P-19 is used for oil skimming TK-5 and the oil is sent through the Rerun line to Rerun Tank TK-100-14.	88-P1-9239	TK 5-0	Process Waste Water Tank
			N/A	Clinton Drive Pipe Interfaces
		88-P1-9250	P-17	Pump
			P-19	Pump
			PM-18A/B	TK-5 Transfer Pumps
46. Global	A review of global scenarios that applies to all nodes in Galena Park			
47. MOC Review	MOC Review for past 5 years for Galena Park Deviation column - MOC # Causes column - Date requested Consequences column - MOC Title Safeguards column - Technical Basis Remark column - MOC Description/Justification			
48. Incident Review	Incident Review for past 5 years for Galena Park Deviation Column - Incident ID # Causes column - Incident Description Consequences column - Date/Time of Incident Remark column - Immediate Action Taken			

The Study team utilized a combination of **HAZOP and What if / Checklist** methodologies to modify any nodes / topics as applicable.

3. Risk Ranking

The risk ranking was evaluated using the current Kinder Morgan Risk Ranking Matrix, shown in Appendix A – Risk Ranking Matrix. The risk matrix was used to qualitatively assess the risk associated with each identified cause / consequence scenario that presented hazardous consequences. For the purpose of this analysis, Hazardous consequences were considered to be those situations that could result in a loss due to injury, environmental impact, or financial loss.

3.1. Assigning Severity and Likelihood

The team first identifies credible worst case consequences, as well as engineering and administrative safeguards. *Severity* is then assigned based on the credible worst case consequences without utilizing mitigation from the engineering or administrative safeguards. Likelihood is then assessed with engineering and administrative safeguards in place. The *severity* is assigned based on a number rating system of '1' to '5'. *Likelihood* is assigned based on a number rating system of '1' to '5'. The descriptions can be found on the risk-ranking matrix in Appendix A – Risk Ranking Matrix.

3.2. Risk Ranking

The descriptions can be found below and on the risk-ranking matrix in Appendix A – Risk Ranking Matrix. Utilizing the risk matrix, the scenario is assigned a Risk Rank value of '1' to '6'. The PHA team shall assign action items as follows:

- Risk Rank 1 - Negligible risk; no action required.